

Process and Measurement System Capability Analysis

Cp & Cpk capability ratios · capability from control charts · gauge (measurement system) capability

Slides covered: 3-4, 10-12, 15-16, 19-25, 28-35

Summary + all formulas & examples

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Process Capability & Natural Tolerance Limits

slides
3-4

Process capability analysis quantifies process variability and analyzes it relative to product specifications. **Process capability** refers to the **uniformity** of the process. Variability is viewed two ways:

1. The natural / inherent variability at a specified time — **“instantaneous” variability**.
2. The variability **over time**.

It is customary to take the **Six-Sigma spread** of the quality-characteristic distribution as the measure of process capability. For a normal distribution with mean μ and standard deviation σ , the **natural tolerance limits** are:

$$\text{UNTL} = \mu + 3\sigma \cdot \text{LNTL} = \mu - 3\sigma$$

Key facts

The natural tolerance limits contain **99.73%** of the output; only **0.27%** falls outside — which is **2,700 nonconforming parts per million (ppm)**. If the output is **non-normal**, the percentage outside $\mu \pm 3\sigma$ may differ considerably from 0.27%.

σ is almost always unknown and is replaced by an estimate $\hat{\sigma}$ (from s or $\bar{R}/d_2$). The **process capability ratio (PCR)** for two-sided specifications:

Cp

$$\hat{C}_p = \frac{USL - LSL}{6 \hat{\sigma}}$$

The PCR has a practical interpretation — the **percentage of the specification band used up** by the process:

$$P = \frac{1}{\hat{C}_p} \times 100$$

One-Sided PCRs

slide 11

$$C_{pu} = \frac{USL - \mu}{3\sigma} \quad (\text{upper only}) \quad \cdot \quad C_{pl} = \frac{\mu - LSL}{3\sigma} \quad (\text{lower only})$$



Example — bursting strength (LSL = 200)

$\bar{x} = 264, s = 32$: $C_{pl} = (264 - 200) / (3 \times 32) = \mathbf{0.67}$.

Fallout: area left of $Z = (200 - 264)/32 = -2 \rightarrow$ about **2.28% defective**
 $\approx \mathbf{22,800 \text{ ppm}}$.

Interpretation (Table 8.2)

For a normal, stable process: **C_p = 1** → **2,700 ppm** (two-sided); **C_p = 1.5** → **4 ppm** (one-sided). Higher C_p ⇒ fewer nonconforming parts.

3 C_{pk} — Actual Capability & Centering

slides 15-16

⚠ Why C_{pk}?

C_p does not take process centering into account — it measures **potential** capability, not actual. C_{pk} takes centering into account and measures **actual** capability:

C_{pk}

$$C_{pk} = \min(C_{pu} , C_{pl})$$

- If **C_{pk} = C_p** → the process is **centered** at the midpoint of the specifications.
- If **C_{pk} < C_p** → the process is **off center**.

Capability Analysis Using a Control Chart

Histograms, probability plots and PCRs summarize performance but don't address **statistical control**. The **control chart should be regarded as the primary technique of process capability analysis** (both attributes and variables charts can be used).



Example — bottle strength (Table 8.5, 20 samples of 5)

$\bar{x}$ chart: $CL = \bar{\bar{x}} = 264.06$; $UCL = \bar{\bar{x}} + A_2\bar{R} = 264.06 + (0.577)(77.3) = \mathbf{308.66}$; $LCL = \mathbf{219.46}$.

R chart: $CL = \bar{R} = 77.3$; $UCL = D_4\bar{R} = (2.115)(77.3) = \mathbf{163.49}$; $LCL = D_3\bar{R} = 0$.

Estimate σ from the range: $\hat{\sigma} = \bar{R}/d_2 = 77.3 / 2.326 = \mathbf{33.23}$. With $LSL = 200$: $C_{pl} = (264.06 - 200)/(3 \times 33.23) \approx \mathbf{0.64}$.

Observed variability comes partly from the units measured and partly from the **measurement system** (instrument/gauge, operator, conditions, calibration). Purpose of a measurement-systems capability study:

1. Determine how much of the observed variability is due to the gauge.
2. Isolate the components of variability in the measurement system.
3. Assess whether the gauge is **capable** (suitable for its intended use).

The two R's of measurement capability

Repeatability — same value when the same unit is measured several times under *identical* conditions. **Reproducibility** — variation when measured under *different* conditions (operators, time...). Other aspects: **linearity, stability, bias.**

Basic MSA model — total observed measurement = true value + measurement error:

Eq 8.23

$$y = x + \varepsilon \implies \sigma^2_{\text{Total}} = \sigma^2_p + \sigma^2_{\text{Gauge}}$$

The Precision-to-Tolerance (P/T) Ratio

Compares gauge capability to the tolerance band (USL – LSL):

$$P/T = \frac{k \cdot \hat{\sigma}_{\text{Gauge}}}{USL - LSL}$$

Popular constants: **k = 5.15** (95% tolerance interval covering $\geq 99\%$ of a normal population) or **k = 6** (the usual natural tolerance limits).



Example

$\hat{\sigma}_{\text{Gauge}} = 0.887$, $USL = 60$, $LSL = 5$, $k = 6$: $P/T = 6(0.887)/(60 - 5) = 5.32/55 = \mathbf{0.097}$.

$P/T \leq 0.1$ is usually taken to imply **adequate gauge capability** (use with caution).

Variance Components, SNR & Discrimination Ratio

From a gauge study, estimate the total standard deviation, then separate the product and gauge parts:

$$\hat{\sigma}_{\text{Total}}^2 = s^2 \cdot \hat{\sigma}_P^2 = \hat{\sigma}_{\text{Total}}^2 - \hat{\sigma}_{\text{Gauge}}^2$$



Example 8.7 (continued)

$s = \sqrt{(392.4/39)} = 3.17 \rightarrow \hat{\sigma}_{\text{Total}}^2 = (3.17)^2 = 10.05$; $\hat{\sigma}_P^2 = (3.17)^2 - (0.887)^2 = 9.26$; $\hat{\sigma}_P = \mathbf{3.04}$.

Ratios of variability

slide 31

$$\rho_P = \frac{\sigma_P^2}{\sigma_{\text{Total}}^2} \quad (8.26) \cdot \rho_M = \frac{\sigma_{\text{Gauge}}^2}{\sigma_{\text{Total}}^2} \quad (8.27) \cdot \rho_P = 1 - \rho_M$$

Example: $\hat{\rho}_M = 0.79/10.05 = \mathbf{0.0786} \rightarrow$ the instrument contributes $\approx \mathbf{7.86\%}$ of total variance, so $\rho_P = 0.9214$.

Signal-to-Noise Ratio (SNR)

slide 32

Eq 8.28

$$\text{SNR} = \sqrt{\frac{2 \rho_P}{1 - \rho_P}}$$

The number of distinct levels reliably obtained. AIAG: **SNR ≥ 5** recommended; **< 2** inadequate. Example: $\text{SNR} = \sqrt{(2 \cdot 0.9214 / (1 - 0.9214))}$

= **4.84** → does *not* meet the $\text{SNR} \geq 5$ guideline.

Discrimination Ratio (DR) slide 33

Eq 8.29

$$\text{DR} = \frac{1 + \rho_p}{1 - \rho_p}$$

For a capable gauge, **DR > 4**. Example: $\text{DR} = (1 + 0.9214)/(1 - 0.9214) = \mathbf{24.45}$ → the gauge is capable by this measure.

8 Accuracy, Precision & Gauge R&R slides 34-35

Accuracy vs. Precision

Accuracy — the ability of the instrument to measure the true value correctly *on average* (evaluated using a known standard; fixed by calibration). **Precision** — the inherent variability in the measurement system.

The gauge variance itself splits into two components — the basis of a **gauge R & R study**:

$$\sigma^2_{\text{Gauge}} = \sigma^2_{\text{Repeatability}} + \sigma^2_{\text{Reproducibility}}$$

Repeatability = inherent precision of the gauge itself; **Reproducibility** = variability due to different operators / conditions.



Formula Cheat Sheet

Quantity	Formula	Note
Natural tolerance limits	$UNTL/LNTL = \mu \pm 3\sigma$	99.73% inside · 2700 ppm out
Process capability ratio	$Cp = (USL - LSL) / (6\sigma)$	two-sided, potential
% of spec band used	$P = (1/Cp) \times 100$	
One-sided PCR	$Cpu = (USL - \mu)/3\sigma$; $Cpl = (\mu - LSL)/3\sigma$	
Actual capability	$Cpk = \min(Cpu, Cpl)$	$Cpk=Cp \rightarrow$ centered
Estimate σ from range	$\hat{\sigma} = \bar{R} / d_2$	variables charts
MSA model	$y = x + \varepsilon$; $\sigma^2_{Total} = \sigma^2_P + \sigma^2_{Gauge}$	
Precision-to-tolerance	$P/T = k \cdot \hat{\sigma}_{Gauge} / (USL - LSL)$	$k=5.15$ or $6 \cdot P/T \leq 0.1$ good
Variability ratios	$\rho_P = \sigma^2_P / \sigma^2_{Total}$; $\rho_M = \sigma^2_{Gauge} / \sigma^2_{Total}$	$\rho_P = 1 - \rho_M$
Signal-to-noise ratio	$SNR = \sqrt{(2\rho_P / (1 - \rho_P))}$	≥ 5 good, < 2 bad
Discrimination ratio	$DR = (1 + \rho_P) / (1 - \rho_P)$	> 4 capable
Gauge R&R	$\sigma^2_{Gauge} = \sigma^2_{Repeatability} + \sigma^2_{Reproducibility}$	



Quick Review

- **Capability** = process uniformity; measured by the **Six-Sigma spread**; natural limits $\mu \pm 3\sigma$ (2700 ppm out).
- **Cp** = $(USL - LSL) / 6\hat{\sigma} \rightarrow$ *potential* capability (ignores centering); **Cpk** = **min(Cpu, Cpl)** $\rightarrow$ *actual* capability. $Cpk < Cp \Rightarrow$ off center.
- The **control chart** is the primary technique of capability analysis; estimate $\hat{\sigma} = \bar{R} / d_2$.
- **MSA**: $\sigma^2_{\text{Total}} = \sigma^2_P + \sigma^2_{\text{Gauge}}$; gauge splits into **repeatability** + **reproducibility** (gauge R&R).
- Gauge adequacy: **P/T $\leq$ 0.1, SNR $\geq$ 5, DR $>$ 4.**